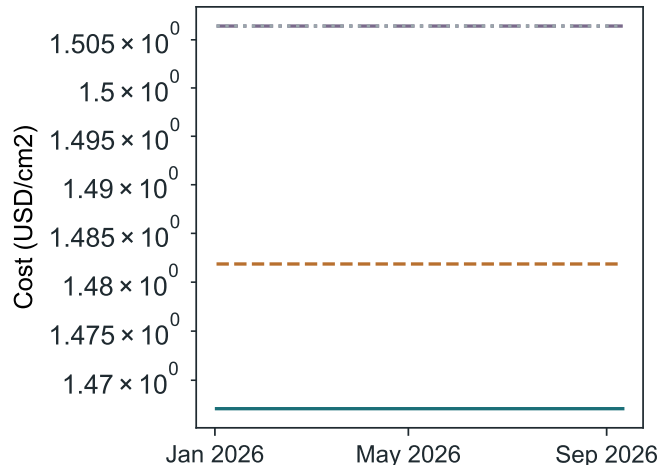
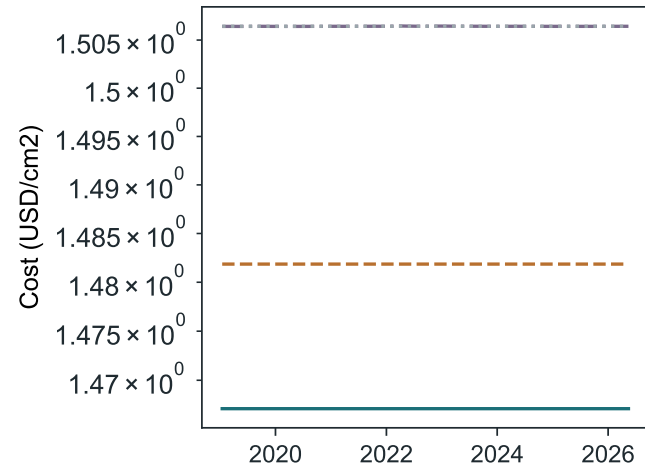


AEM electrolyzer OER

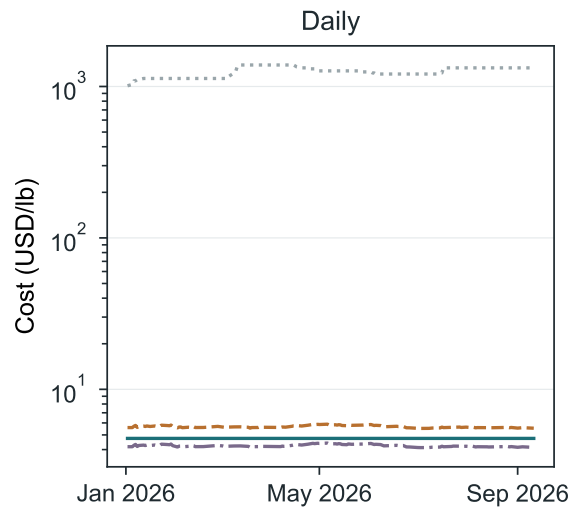
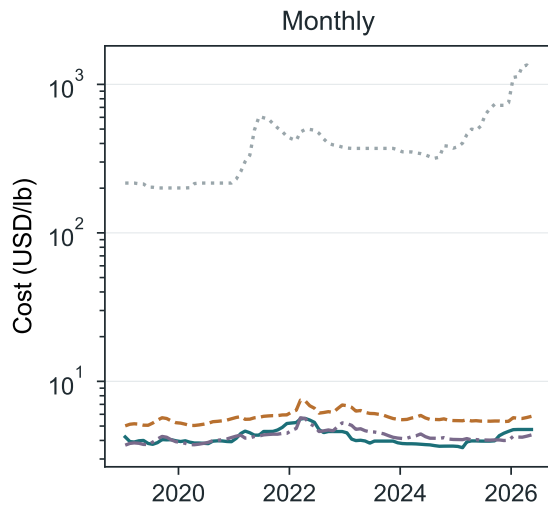
Monthly

Daily



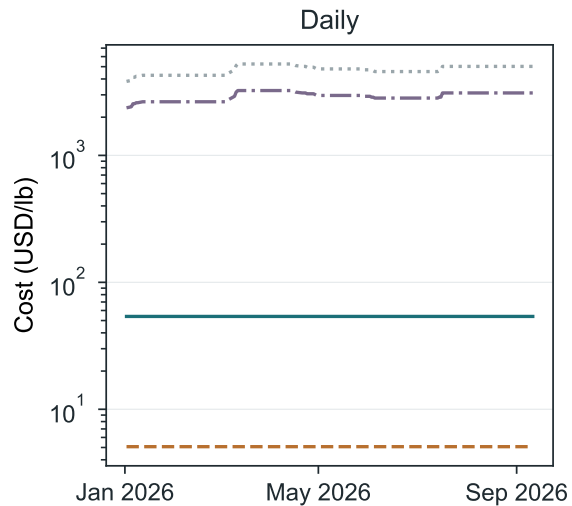
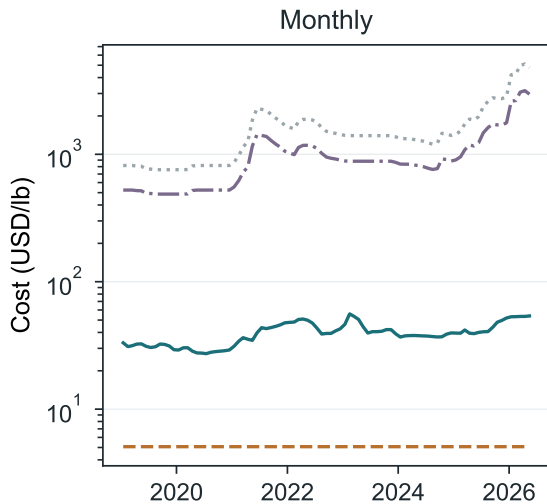
- cr-doped-amorphous
- nife-seed-assisted
- nico2o4-spinel
- nife-lah-scalable

Ammonia cracking



- co-mgo-la2o3
- ni-mgo-ceo2-interface
- ni-alumina-baseline
- ru-mgo-premium

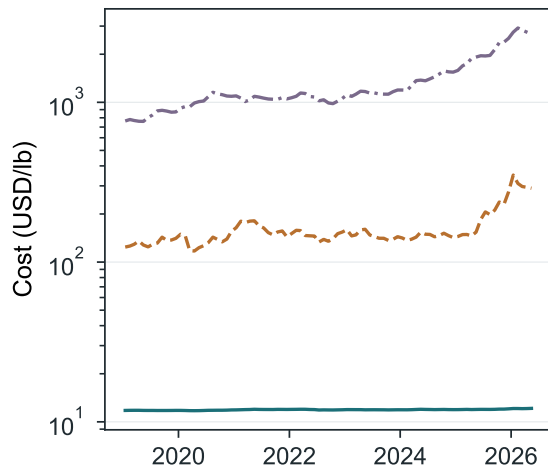
Ammonia synthesis



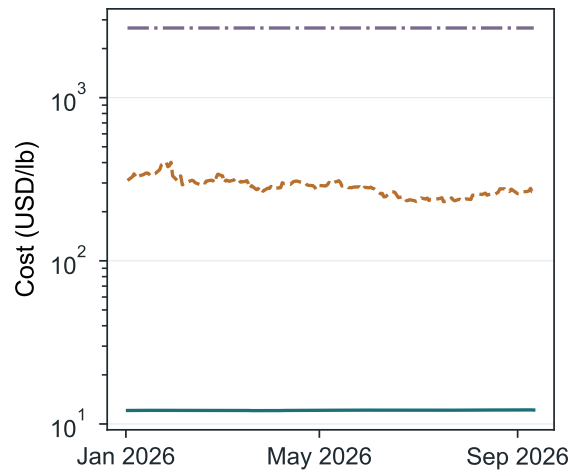
- cs-co3mo3n
- fused-iron-baseline
- ru-graphite-kaap
- ru-cafh-lowtemp

Preferential CO oxidation

Monthly



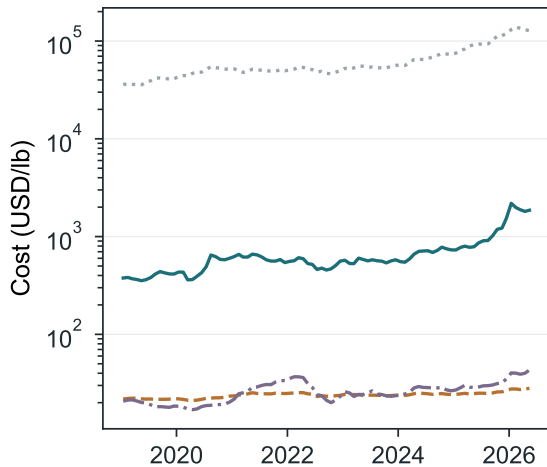
Daily



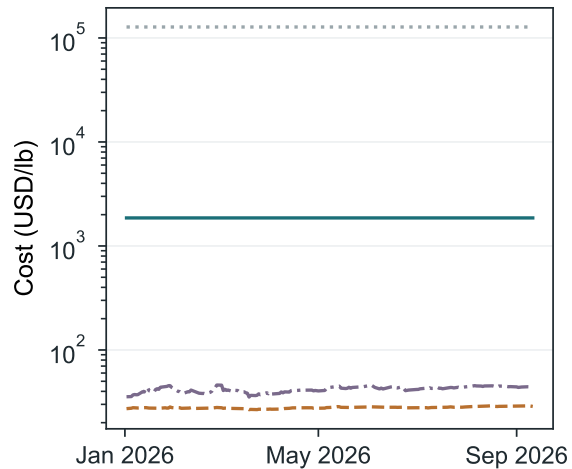
- cuo-ceo2
- - - pt-fe-alumina
- . - au-tio2-nanogold

CO₂ electroreduction

Monthly



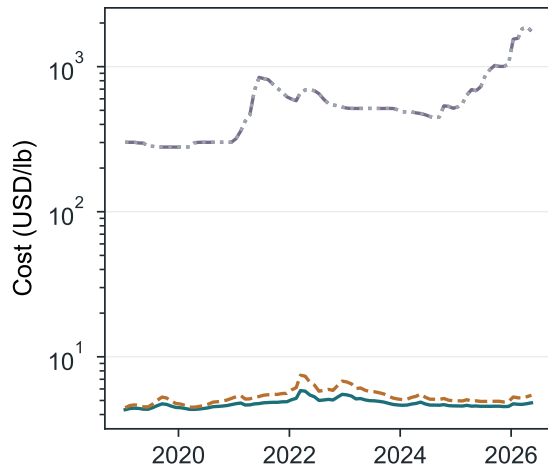
Daily



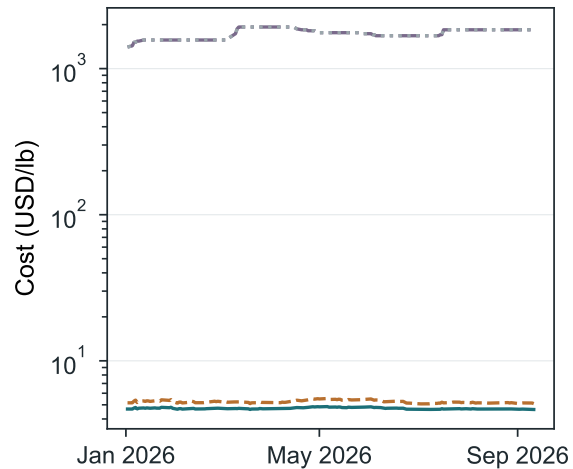
- ag-co-mea
- - - cu-multicarbon-co2rr
- . - sn-formate-co2rr
- au-co-co2rr

CO₂ methanation

Monthly



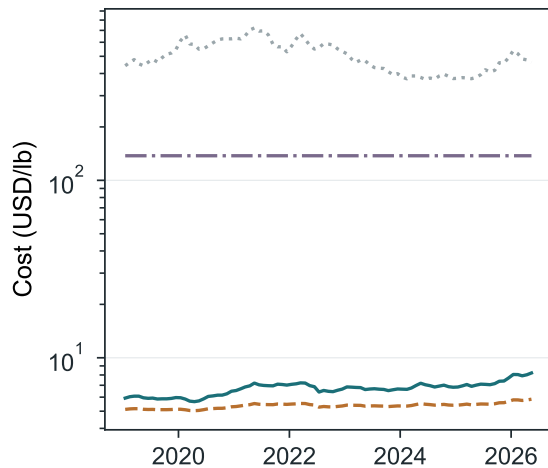
Daily



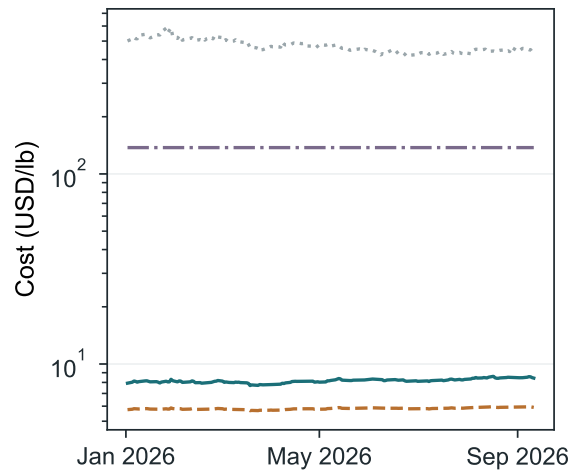
- ni-ceo2-lowtemp
- ni-alumina-baseline
- ru-layered-titanate
- ru-mnox-photothermal

CO₂ to methanol

Monthly

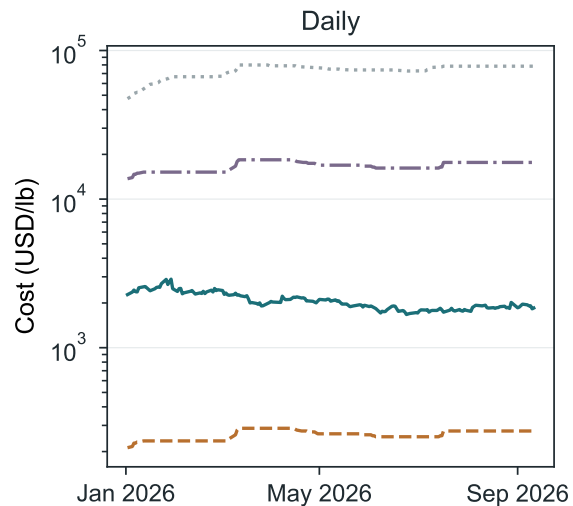
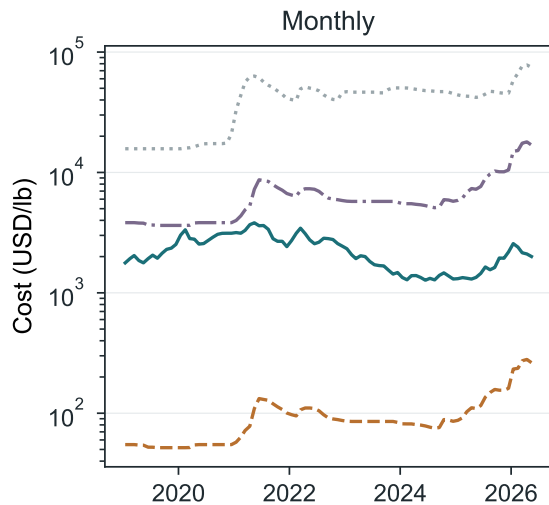


Daily



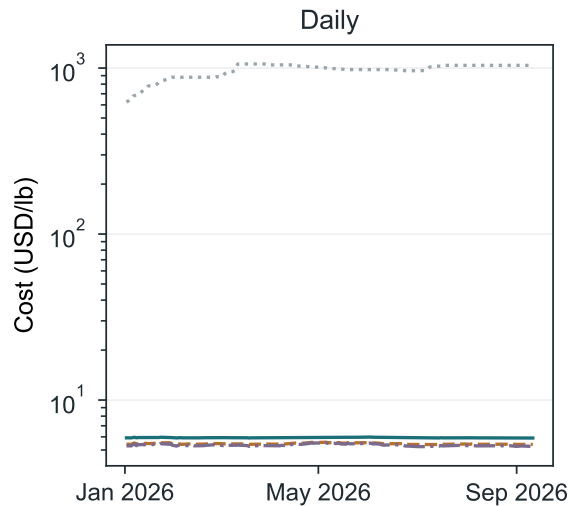
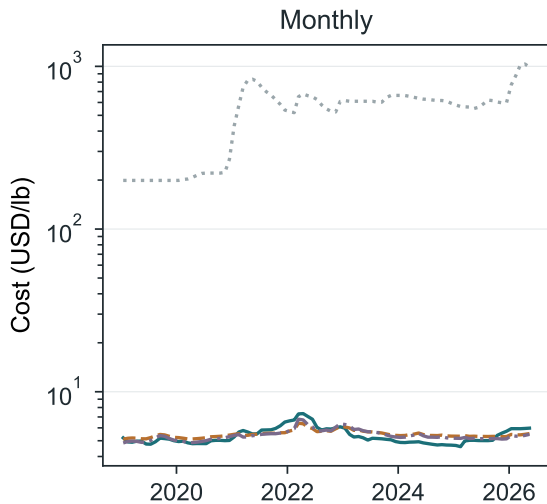
- cza-baseline
- - - cu-zrox-interface
- . - in2o3-zro2-lowtemp
- pd-in2o3

CO₂ to formate



- pdag-ndcarbon-co2fa
- ru-sa-ldh
- ru-pincer-co2fa
- ir-pincer-co2fa

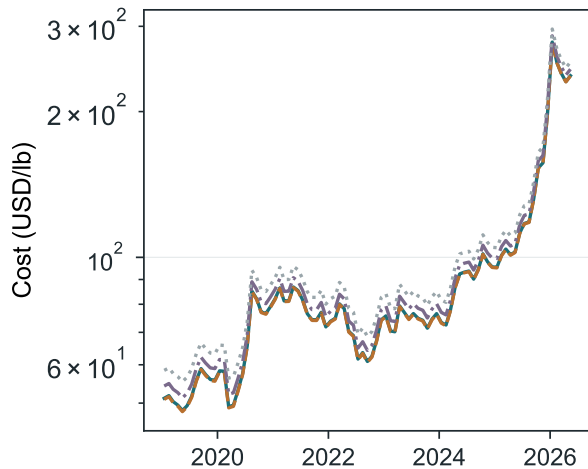
Methane dry reforming



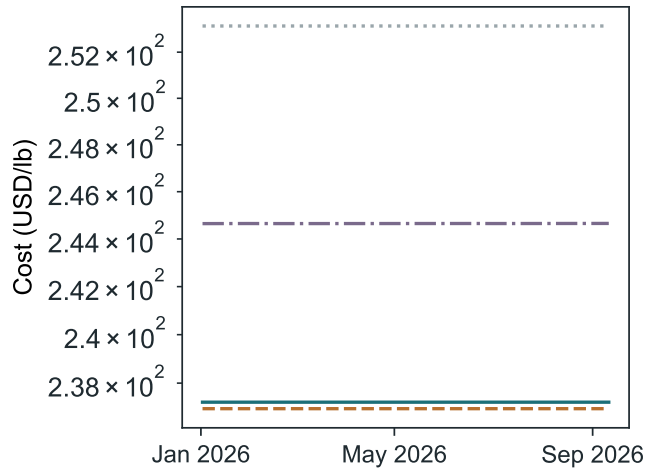
- ni-co-almgo
- ni-single-atom-ceria
- ni-zeolite-stable
- ir-ceria-premium

Ethylene epoxidation

Monthly

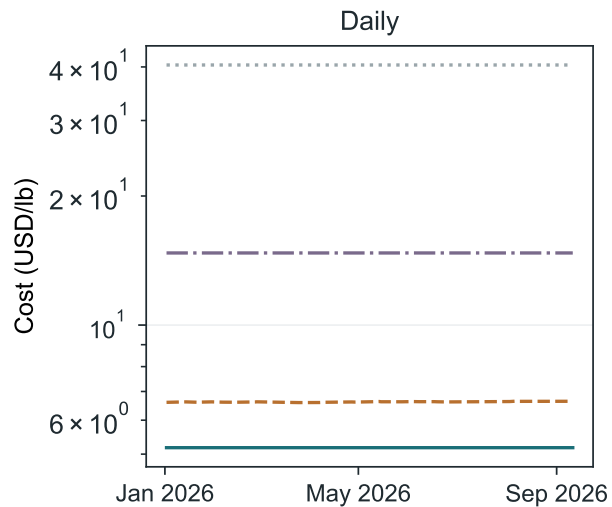
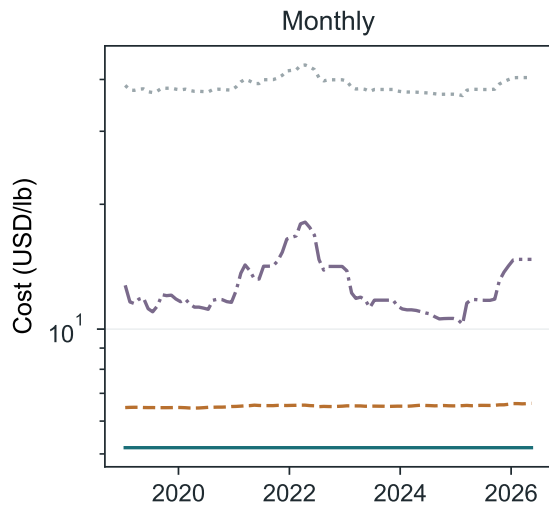


Daily



- agcs-industrial
- - ag-alumina-baseline
- . cu-ag-bimetallic
- . . . agcsre-modern

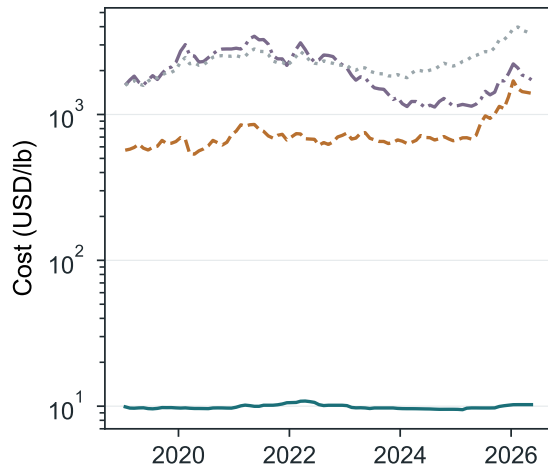
Fischer–Tropsch synthesis



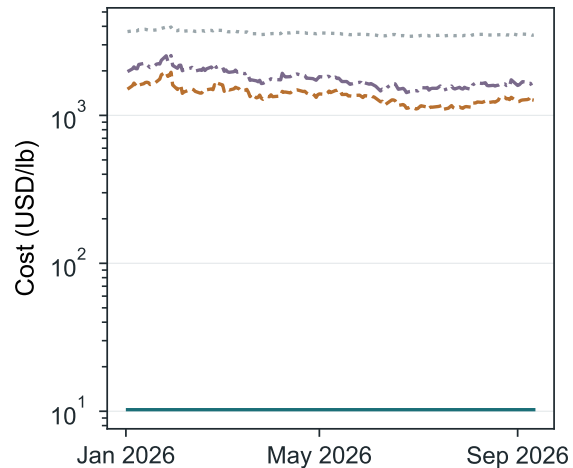
- fe-silica-lower-olefin
- fe-cu-k-precipitated
- co-alumina-fts
- co-cnt-structured

Formic acid dehydrogenation

Monthly



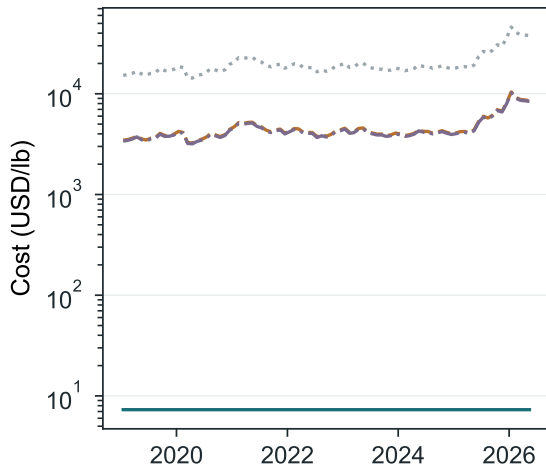
Daily



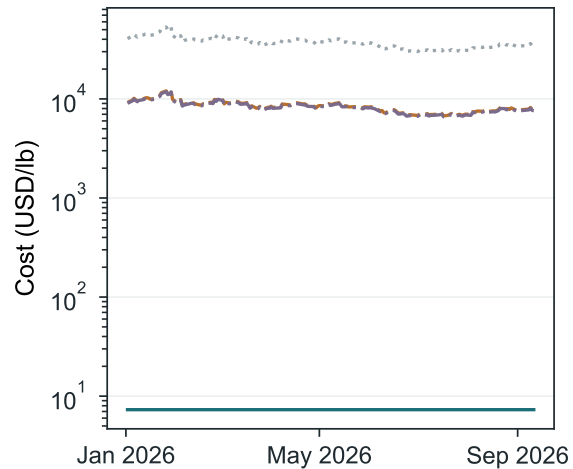
- co-nc-nonnoble
- ptmo-gasphase-premium
- pd-carbon-baseline
- aupd-mil101

Fuel-cell ORR

Monthly



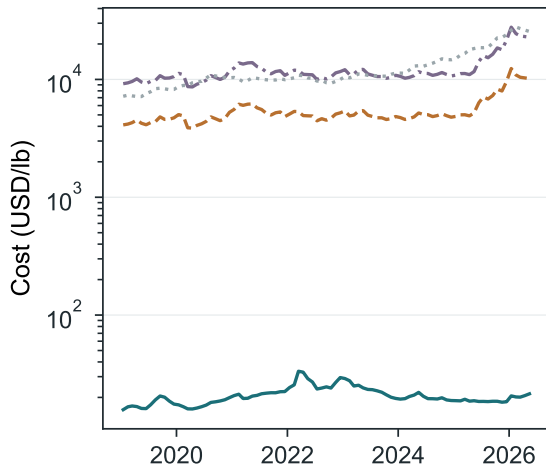
Daily



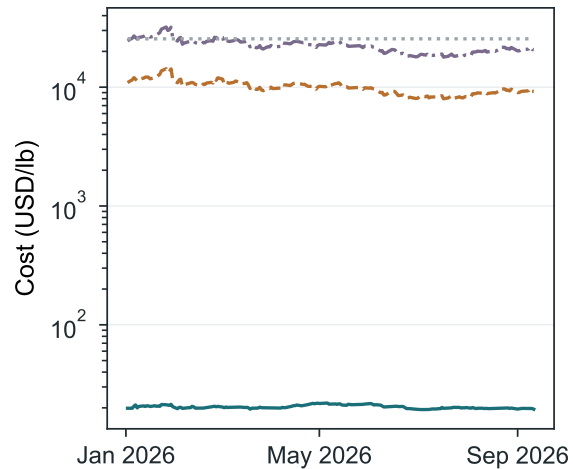
- fen-c-pgmfree-cathode
- ptco-intermetallic-cathode
- ptc-baseline-cathode
- ptni-octahedra

Glycerol electro-oxidation

Monthly



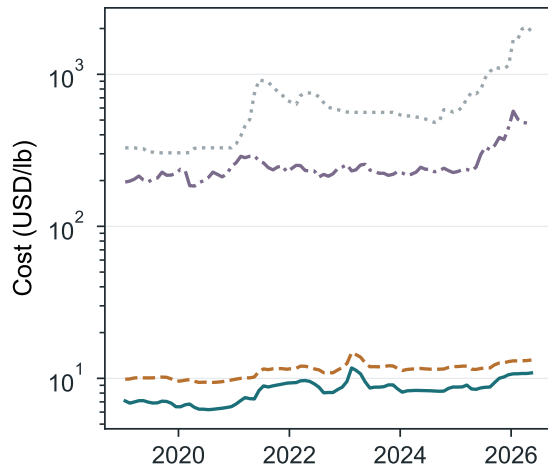
Daily



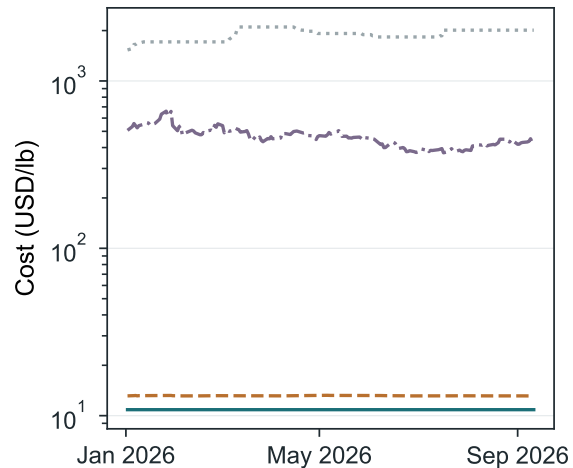
- niooh-glycerol-anode
- pt-c-glycerol-anode
- pt-bi-dha
- au-c-glycerol-selectivity

Bio-oil hydrodeoxygenation

Monthly

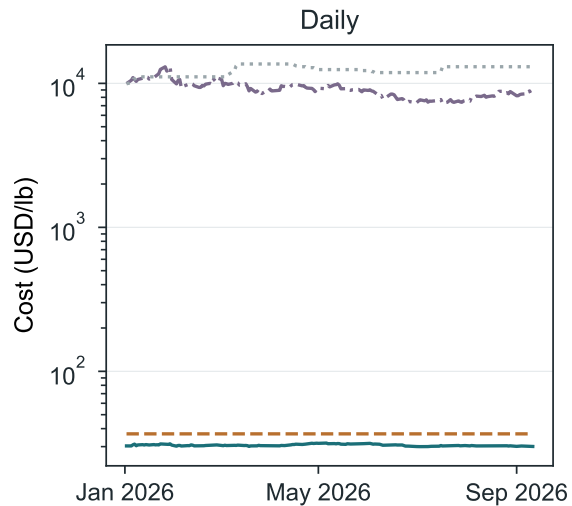
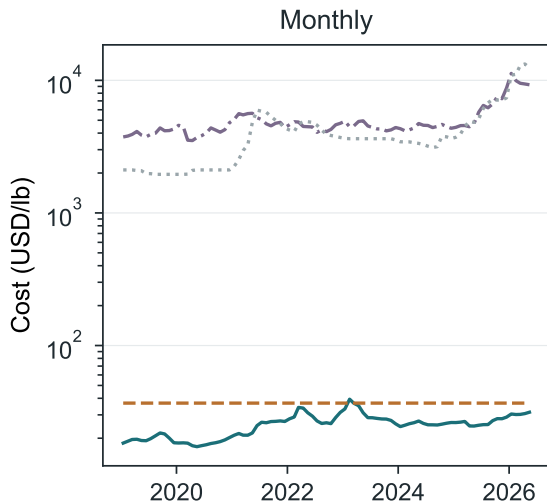


Daily



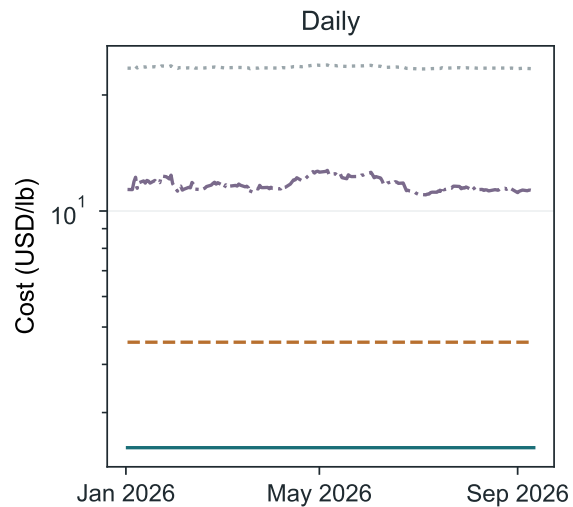
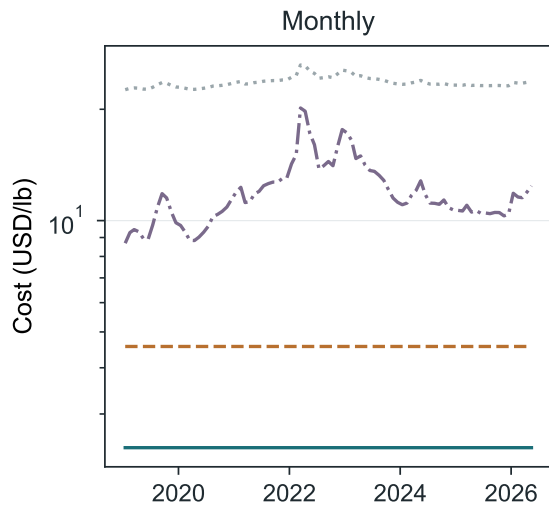
- comos-hdo
- nimo-carbon-hdo
- pt-silica-hdo
- ru-carbon-hdo

Hydrogen evolution reaction



- nimo-alkaline-her-cathode
- mos2-acidic-her-cathode
- pt-carbon-her-cathode
- ru-c2n-her

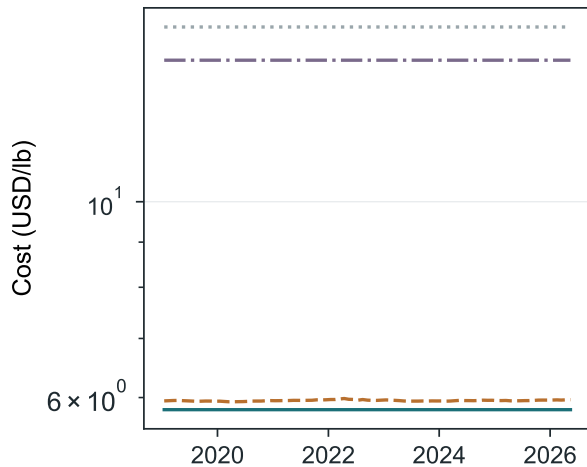
Methane pyrolysis



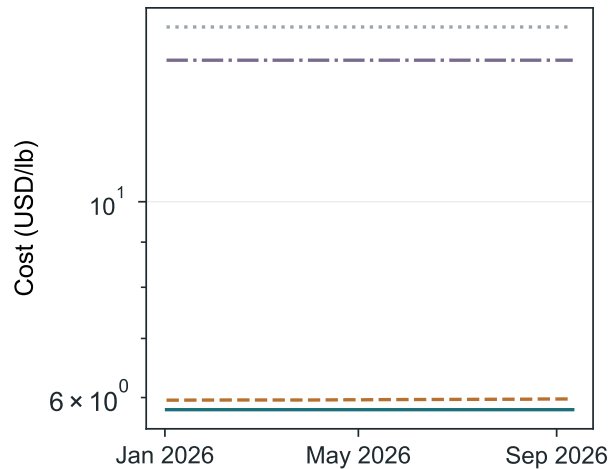
- activated-carbon-autocatalytic
- - - fe-alumina-pyrolysis
- . - ni-sio2-highload
- molten-ni-bi

Methanol to olefins

Monthly

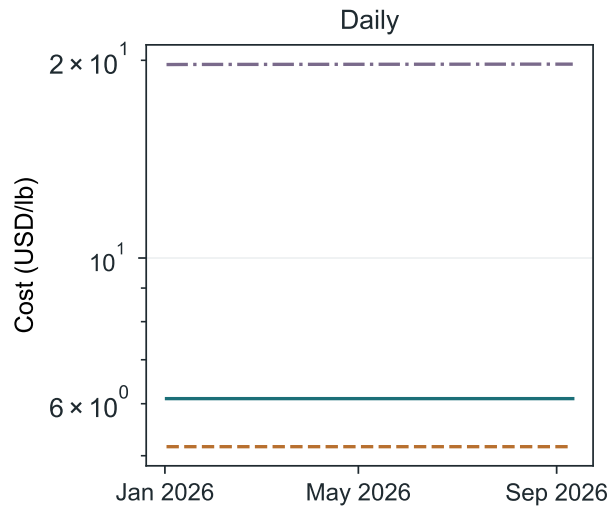
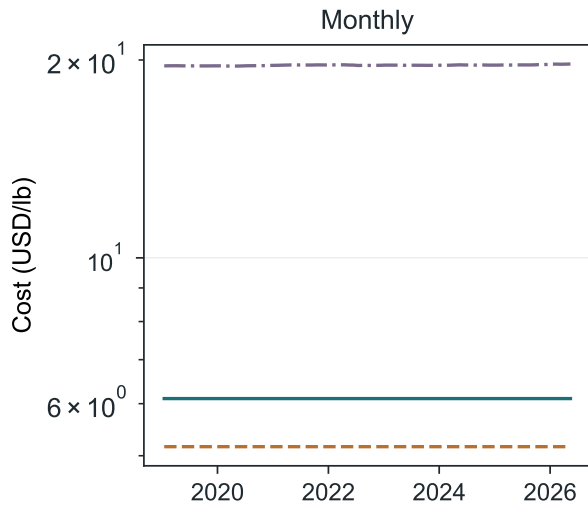


Daily



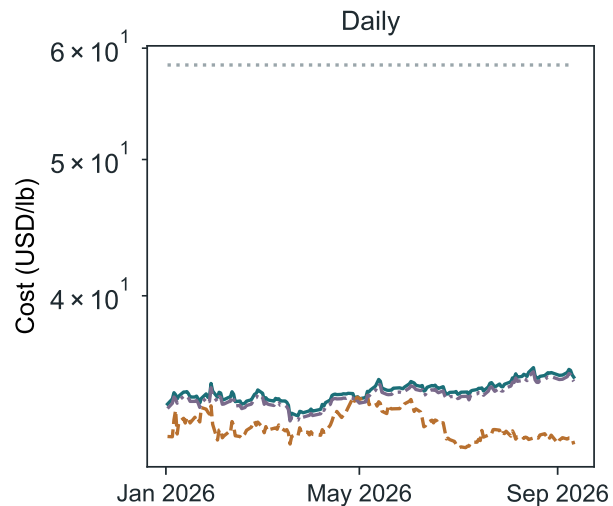
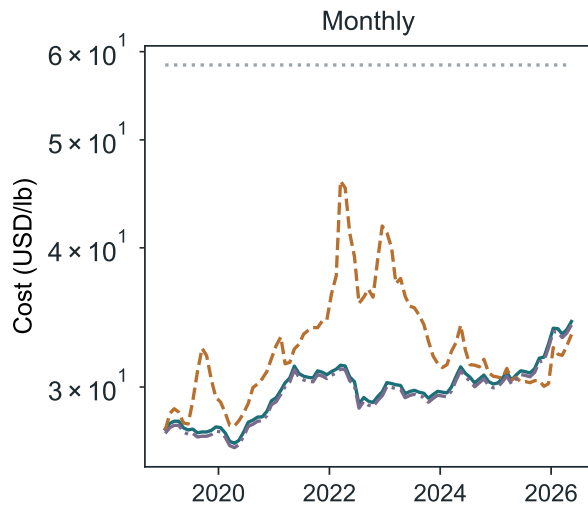
- hzsm5-mth
- zn-zsm5-aromatics
- sapo-34-light-olefin
- nano-sapo-34

NH₃-SCR



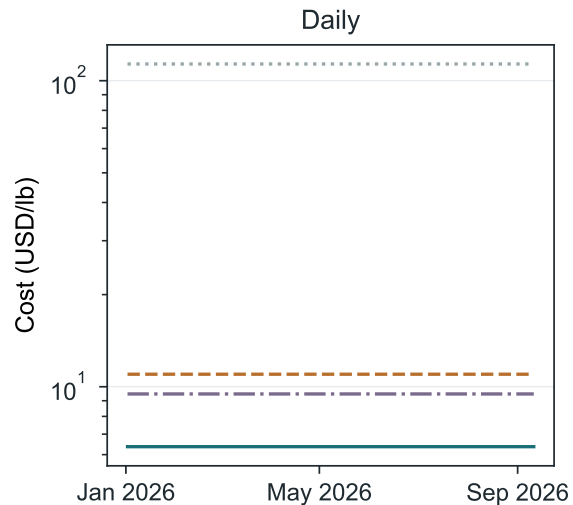
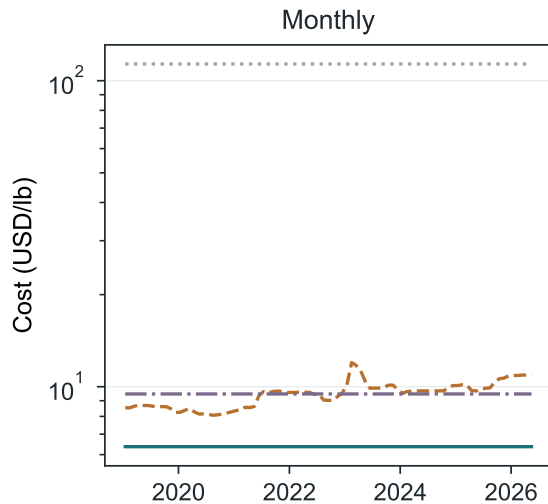
— v2o5-wo3-tio2
- - fe-zsm-5
- . cu-ssz-13

Electrochemical N₂ reduction



- li-mediated-nrr
- plasma-nrr
- aqueous-nrr-cu
- li-flow-shuttle

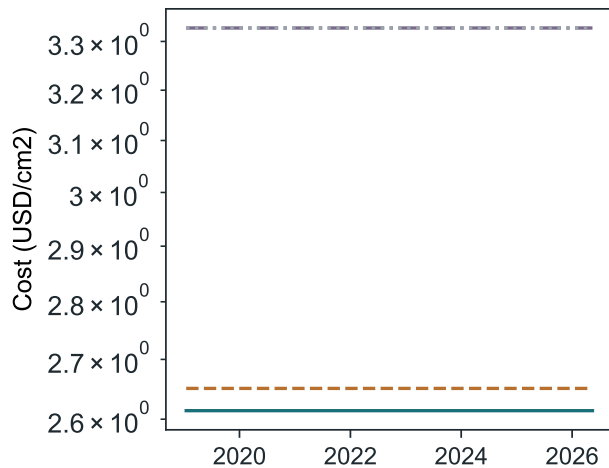
Olefin metathesis



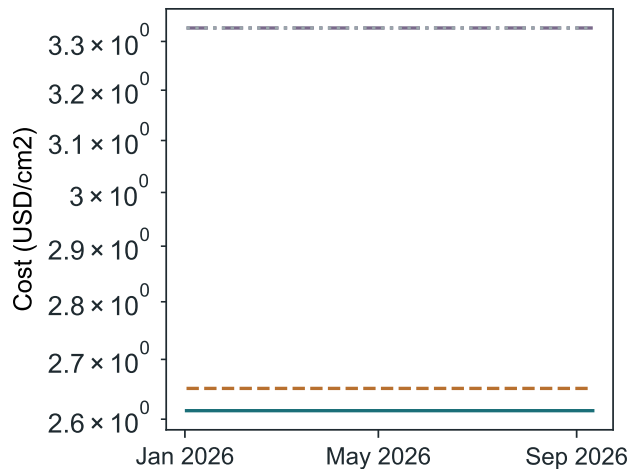
- wo3-silica-oct
- mo-silica-alumina
- w-h-alumina
- re2o7-alumina-mild

PEM electrolyzer OER

Monthly



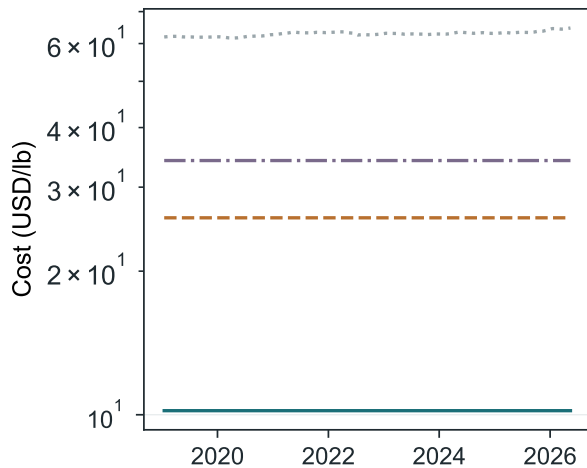
Daily



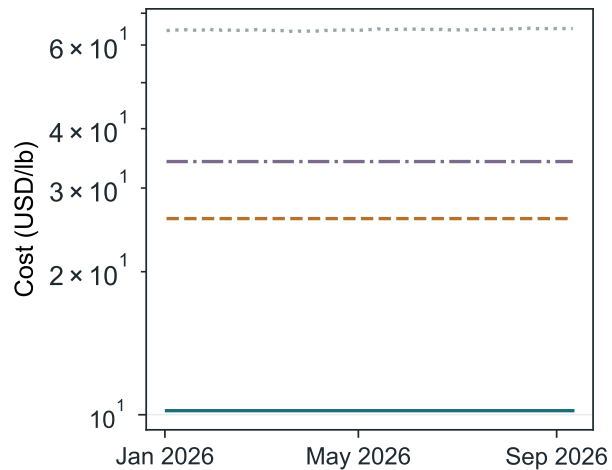
- pem-low-ir-interface
- - pem-ru-rich-acidic
- . iro2-tio2-supported
- pem-irox-baseline

Photocatalytic CO₂ reduction

Monthly

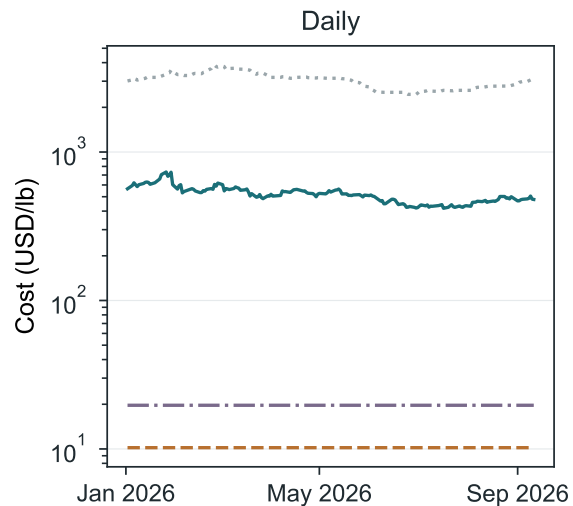
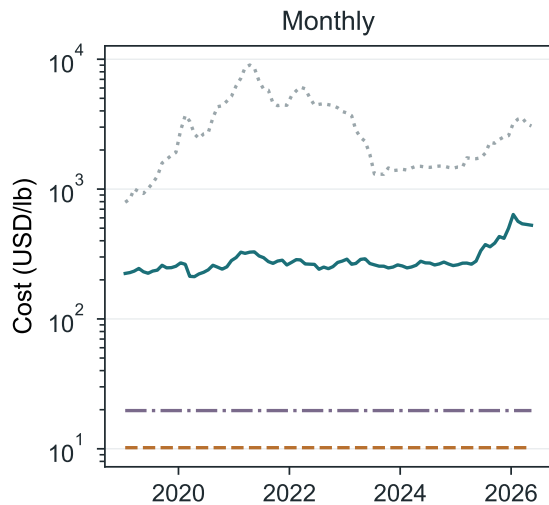


Daily



- tio2-photo-co2
- gcn-sns2-zscheme
- bivo4-visible
- cu2o-mof-co2

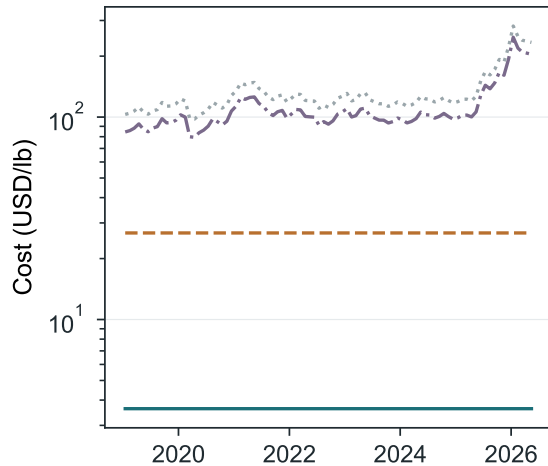
Photocatalytic water splitting



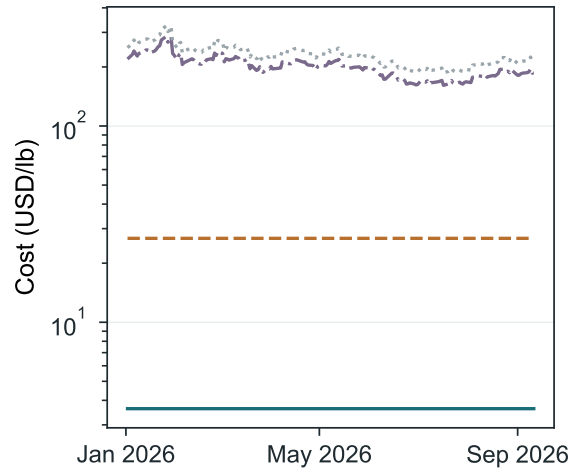
- pt-tio2-cocatalyst
- tio2-anatase-baseline
- gcn-metal-free
- sr tio3-al-qe-benchmark

Propane dehydrogenation

Monthly

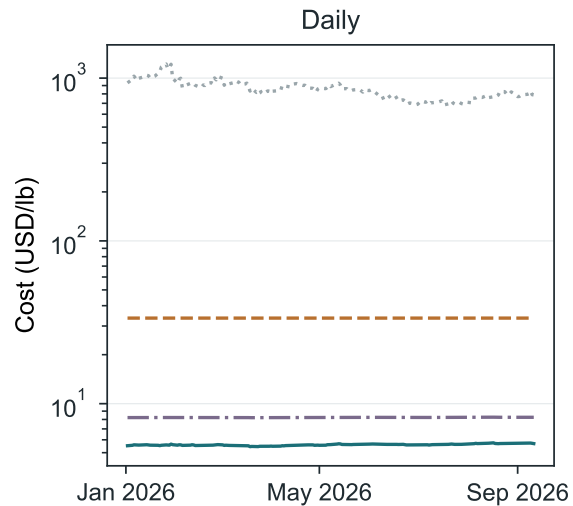
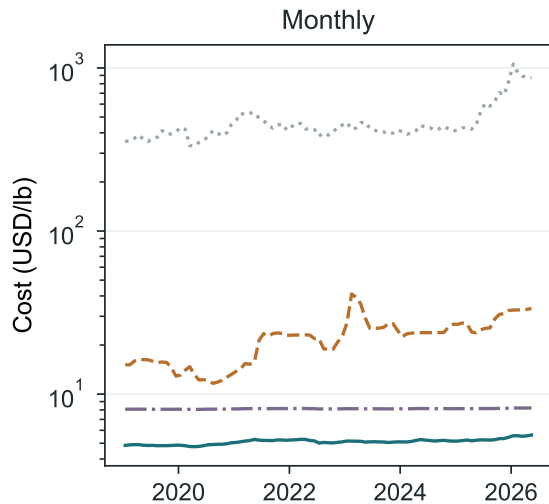


Daily



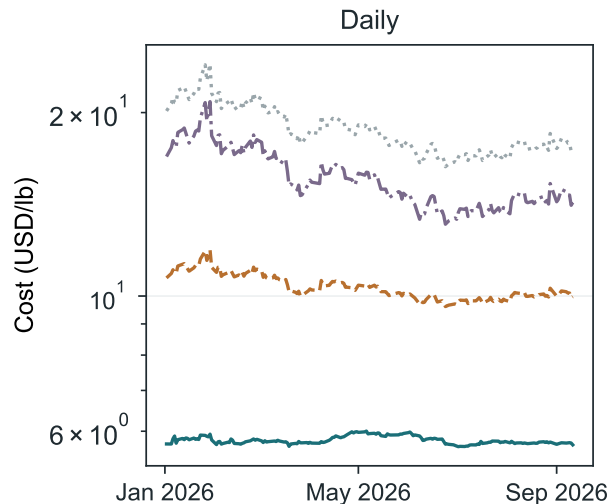
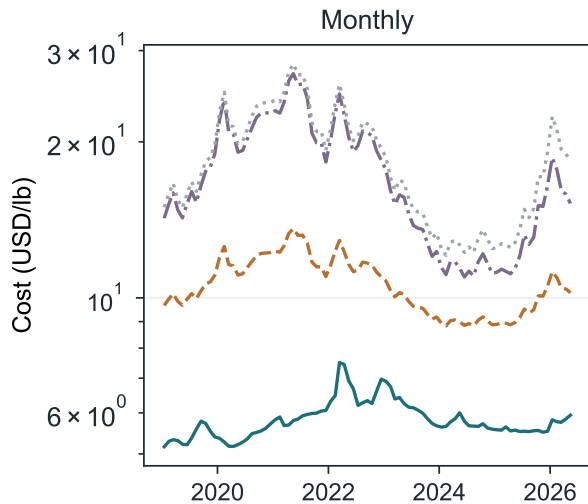
- crox-alumina-catofin
- - - bn-odhp
- . - ptsn-alumina-pdh
- ptzn-intermetallic-pdh

Reverse water-gas shift



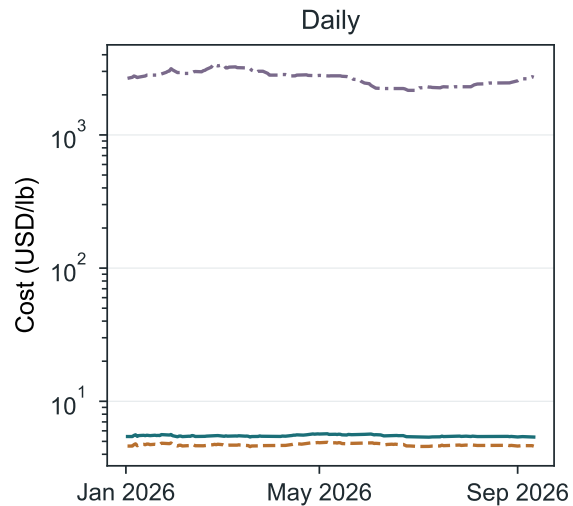
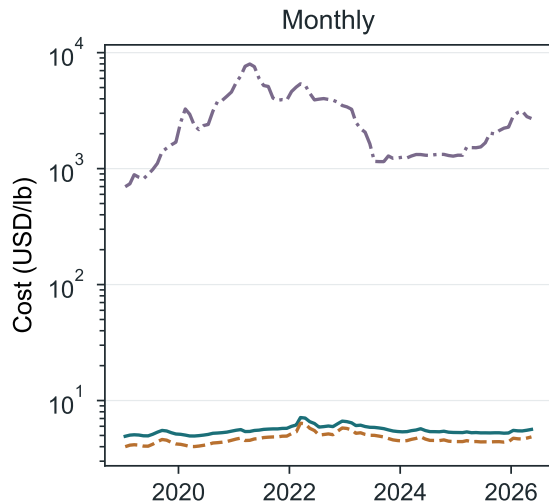
- cu-ceria-baseline
- - - mo2n-hightemp
- . - cs-fe-cu-alumina
- pt-tio2-premium

Selective acetylene hydrogenation



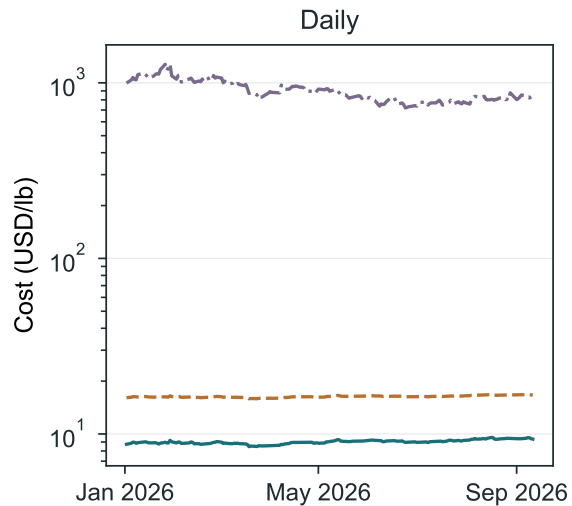
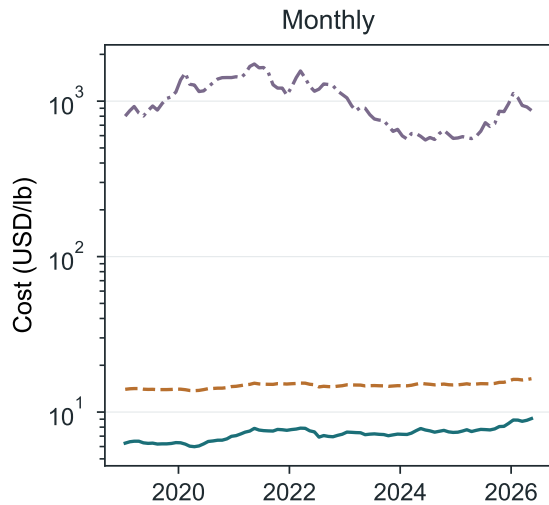
- nizn-intermetallic
- pdcu-saa
- pd-alumina-baseline
- pdag-bimetallic

Steam methane reforming



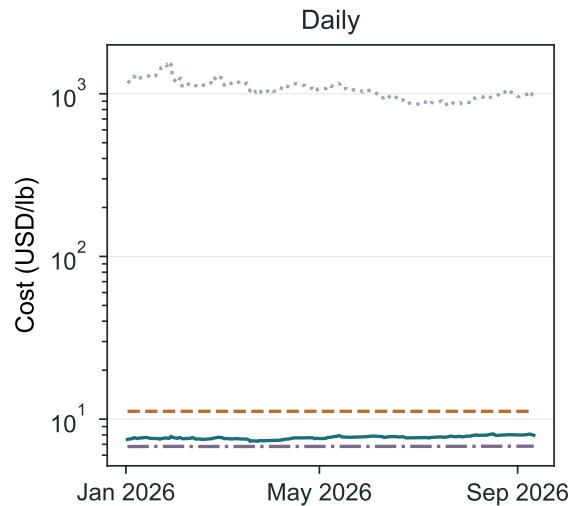
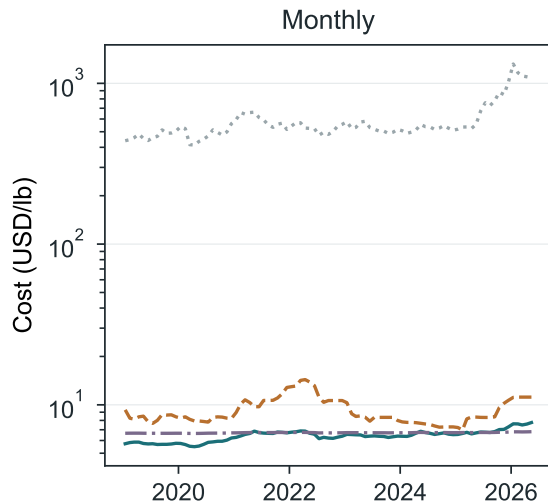
- ni-calcium-aluminate
- sn-ni-coking-resistant
- rh-alumina-compact

Syngas to methanol



- cza-industrial
- - ga-promoted-cza
- . - pd-zno-intermetallic

Water-gas shift



- cu-zno-baseline
- - co-ceria-interface
- . fe-cr-hts
- pt-ceria-premium